



Review

Comprehensive review of methodologies and case studies in advanced exergy, exergo-economic, and exergo-environmental analyses

M. Fallah^{a,*}, Z. Mohammadi^b, S. Allahyari^c, SH. Rahimi^d, M. Fathi^e, Z. Hoseini Tabar^f, S.M.S. Mahmoudi^g^a Research Institute of Applied Power System Studies, Azarbaijan Shahid Madani University, Tabriz, Iran^b Engineering Faculty, Azarbaijan Shahid Madani University, Tabriz, Iran^c School of Mechanical Engineering, Iran University of Science and Technology, Tehran, Iran^d Mork Family Department of Chemical Engineering and Material Science, University of Southern California, Los Angeles, USA^e School of New Technologies, Iran University of Science and Technology, Tehran, Iran^f Mechanical Engineering Department, Islamic Azad University, Hamedan Branch, Hamedan, Iran^g Mechanical Engineering Faculty, Tabriz University, Tabriz, Iran

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ABSTRACT

Advanced exergy analysis is an effective method for evaluating energy systems' sustainability and efficiency. In contrast to conventional exergy analysis, which finds and measures the irreversibilities in a system, advanced exergy analysis goes farther and makes a distinction between exergy losses and destructions that can be prevented and those that cannot. This distinction gives engineers important information about the real potential for energy system improvement, enabling them to choose changes that will result in the biggest efficiency increases. By considering the interactions between system components, the approach emphasizes the influence of component performance on the overall system. Advanced exergo-economic analysis provides a more precise assessment by integrating economic factors with exergy destructions, enabling detailed cost identification for both avoidable and unavoidable exergy destruction. Similarly, advanced exergo-environmental analysis offers a refined approach by accurately tracing environmental impacts within energy systems, distinguishing between manageable and inherent emissions. This work also investigates current developments in the subject, such as the use of advanced exergy analysis to support sustainable development goals and its integration with economic and environmental methodologies. This essay emphasizes the vital role that advanced exergy analysis will play in advancing sustainability and energy efficiency in the future through a thorough review of case studies and recent research trends. Additionally, a review of previous studies has been conducted, examining advanced exergy, economic, and environmental analysis techniques aimed at achieving higher-efficiency systems.

Nomenclature

AA-CAES	Advanced adiabatic compressed air energy storage	LCES	Liquid carbon dioxide energy storage
AHTHP	Auto cascade high-temperature heat pump	LT-CCES	Low-temperature compressed carbon dioxide energy storage
AV	Avoidable	LTR	Low temperature recuperator
$\dot{B}$	Environmental impact	MARC	Modified auto-cascade Refrigeration cycle
$\dot{C}$	Exergy cost rate (\$/h)	MEA	Modified exergy analysis
CCES	Compressed carbon dioxide energy storage	MOPSA	Modified productive structure analysis

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CCHP	Combined cooling, heating, and power	NSGA-II	Non-dominated sorting genetic algorithm
CCS	CO ₂ capture and storage	ODP	Ozone depletion potential
COP	Coefficient of performance	ORC-CHP	Organic Rankine cycle -combined heat and power
CtEG	Coal to ethylene glycol	ORFC	Organic Rankine flash cycle
d	Destruction	PEMFC	Proton exchange membrane fuel cell
DECHS	Double-efficient CO ₂ heating system	PTES	Pump thermal electricity storage
DORC	Dual-loop organic rankine cycle	$\dot{R}_{tot}$	Total cost rate (\$/h)

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* Corresponding author.

E-mail addresses: mfallah@azaruniv.ac.ir, mf.fallahmohsen@gmail.com (M. Fallah).<https://doi.org/10.1016/j.energy.2025.137606>

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DWH	Domestic water heater	RO	Reverse osmosis
$\dot{E}$	Exergy rate (W)	RORC	Recuperative organic Rankine cycle
ECM	Equivalent component method	SEHSS	Solar-driven ejector heat storage subsystem
EN	Endogenous	SOFC	Solid oxide fuel cells
EX	Exogenous	UN	Unavoidable
EIs	Environmental impact rate (pt/h)	WD	Wall tile dryer
f	Exergo-economic factor	WSH-MES	Wind-solar-hydrogen multi-energy supply
GD	Ground tile dryer	y	Exergy destruction ratio
HRSGs	Heat recovery steam generator	$\dot{Y}$	Environmental impact of a component (\$/h)
HTR	High temperature recuperator	$\dot{Z}$	Investment cost rate (\$/h)
ISCC	Integrated solar combined cycle	ε	Exergy efficiency (%)
LCA	Life cycle assessment		

1. Introduction

Due to the current era's rapid industrialization and urbanization, energy consumption has surged to previously unheard-of levels, worsening the environment and increasing greenhouse gas emissions. New ways to analyze energy that go beyond standard efficiency measurements can make energy systems a lot better. But you should know that sustainable development is a complicated process that involves more than just economic, technological, and environmental concerns. Political, social, and institutional factors are also quite important [1]. Exergy analysis, exergoeconomic evaluation, and exergoenvironmental assessment are sophisticated techniques that could help us learn more about how systems don't perform well and how they harm the environment. They are excellent for the environment and help you manage energy, but they aren't adequate on their own. Exergy analysis, a potent thermodynamic technique to provide a more thorough evaluation of energy systems by considering both the quantity and quality of energy, has drawn a lot of interest. Sophisticated and exact techniques, such as advanced exergy investigation, advanced exergoeconomic evaluation, and exergo-environmental analysis, are now addressing the economic and environmental aspects of energy use. Although conventional exergy analysis has been crucial in identifying system inefficiencies, these advanced techniques offer more profound understandings of systemic inefficiencies, financial consequences, and environmental effects, making them indispensable instruments for sustainable energy management.

Offering thorough insights into the efficiency, economic viability, and ecological sustainability of energy systems, advanced exergy evaluation, advanced exergoeconomic analysis, and exergo-environmental analysis represent important developments in the field of exergy analysis. These techniques provide a better understanding of the reasons behind inefficiencies, the financial effects, and the environmental effects of energy system operation. This review will cover the concepts and uses of advanced exergy analysis, advanced exergoeconomic analysis, and exergo-environmental analysis, along with the various techniques for performing advanced exergy analysis. This review seeks to give a thorough overview of these approaches' responsibilities in increasing environmental sustainability, cost-effectiveness, and energy efficiency by analyzing how they contribute to the understanding and enhancement of energy systems. The insights gained from these sophisticated assessments will greatly aid future energy system development as the energy landscape continues to change.

1.1. Advanced exergy analysis

It is important to distinguish between energy losses (which refer to the quantity of energy not utilized) and exergy destruction, which refers to the loss of energy quality due to irreversibilities in real processes. For

a long time, conventional exergy analysis has been used to find where and how much irreversibility there is in energy systems. However, it often fails to recognize the mutual influence and contribution of individual system components to the overall inefficiencies. To address these constraints, sophisticated exergy analysis has been formulated, providing enhanced understanding by differentiating between endogenous and exogenous, as well as avoidable and unavoidable components of exergy degradation. Kotas [2] was one of the first to talk about them. Tsatsaronis and colleagues [3,4] eventually codified and widely used these ideas in the context of advanced exergy analysis. This improved paradigm has made evaluations more useful by differentiating irreversibilities that are caused by interactions between system components from those that are caused by internal processes. It also helps figure out which aspects of the destruction may actually be averted, which helps prioritize measures for improvement. The concept has now been refined and effectively implemented across several thermal systems, including power plants, refrigeration systems, and chemical processes [3–6]. Morosuk and Tsatsaronis performed sophisticated exergy evaluations on absorption refrigeration cycles employing hybrid methodologies [7], whereas Kelly et al. [8] utilized engineering curve-fitting to identify endogenous components. Researchers assessed the precision of various analytical methodologies, including the thermodynamic cycle method, mass balance method, exergy balance method, equivalent component method, and structural theory method (algebraic notation) VALERO, A., AND TORRES, C. Algebraic thermodynamic analysis of energy systems. *Proceedings of approaches to the design and optimization of thermal systems AES 7* (1988), 13–23. [4,7–10]. Comparative investigations indicate that each strategy yields slightly divergent results contingent upon foundational assumptions and system configurations. The engineering method has, in certain instances, produced slightly elevated efficiency figures compared to the thermodynamic cycle-based method [5]. Some uses of the structural theory technique have showed big differences, but this is usually because the method is sensitive to the data input and because the modeling is too simple. Importantly, contrary to previous misunderstandings, structural theory has been effectively implemented in real-world systems, including actual power plants, as evidenced by the research of Valero et al. [11] Advanced exergy analysis offers a more detailed perspective on irreversibilities by categorizing exergy destruction into endogenous/exogenous and avoidable/unavoidable components. Endogenous destruction occurs due to inherent inefficiencies inside components, whereas exogenous destruction arises from interactions with external components. In the same way, design or operational adjustments might lower avoidable exergy destruction, but unavoidable destruction is thought to be a natural part of present technology or physical restrictions [6,9,12]. These differences assist find places where actual improvements can be made and help optimize the system. Recent studies show that improved exergy methods are now necessary for finding inefficiencies, deciding which changes to make first, and making energy systems more sustainable, cost-effective, and efficient [12–15].

1.2. Advanced exergo-economic analysis

Exergy analysis is a powerful tool for understanding energy efficiency; however, it often overlooks the financial implications of inefficiencies. Exergoeconomics, a subset of thermoeconomics, integrates economic principles with exergy analysis to evaluate the costs associated with exergy destruction and the investments required for system improvements [16]. Conventional exergy-based analyses, including conventional exergetic, exergoeconomic, and exergoenvironmental assessments, exhibit limitations in two crucial areas: component interactions and improvement potential [16,17].

Firstly, conventional analyses inadequately capture the interactions and interrelations among various components within an energy system. These interactions can significantly influence overall system performance, leading to suboptimal efficiency. Secondly, such analyses often

fail to identify the true potential for improving system efficiency, which is essential for optimizing complex energy plants.

To address these limitations, advanced exergy-based analyses have been developed, with a particular emphasis on advanced exergoeconomic analysis. This innovative approach differentiates between avoidable and unavoidable costs, as well as endogenous costs (related to the operation of individual components) and exogenous costs (related to interactions between components). By identifying and characterizing these costs, advanced exergoeconomic analysis enhances cost-effectiveness in thermodynamic systems and provides strategic insights for reducing costs and optimizing performance [17]. This methodology serves as a crucial tool for identifying inefficiencies and guiding improvements, ultimately saving time and resources in system optimization.

This approach has been successfully applied across various industries thermal process [18], including manufacturing, desalination, cooling systems, and power generation, hydrogen liquefaction plants [19] to optimize the balance between capital investments and operating expenses [10,20–22].

1.3. Advanced exergo-environmental analysis

As concerns about environmental sustainability grow, it is critical to assess systems' environmental impact in addition to their energy efficiency and financial performance. To calculate the environmental costs connected to exergy degradation, exergo-environmental analysis integrates exergy analysis and environmental impact assessment. By identifying the elements that contribute most to environmental deterioration, this method helps to focus efforts on lowering emissions and minimizing the ecological footprint of energy systems. This idea is built upon by advanced exergo-environmental analysis, which adds the differences between endogenous and external exergy degradation as well as avoidable and unavoidable exergy destruction [17,20,21,23]. It is feasible to determine which energy system components have the most environmental impact and replace them with more energy-efficient components to lower atmospheric pollution by conducting an advanced exergo-economic study. Advanced exergo-environmental analysis, can pinpoint the main environmental impact sources in energy systems, directing the enhancement of eco-friendlier technologies [5,7].

2. Evaluating conventional versus advanced exergy analysis techniques

In the realm of thermodynamic analysis, exergy analysis has emerged as a vital method for assessing the efficiency of energy systems. Conventional exergy analysis primarily concentrates on pinpointing the locations and amounts of irreversibilities within a system, offering insight into where system improvements may be required. However, conventional exergy destruction cannot identify real opportunity enhancement and interaction between the system components.

Conversely, advanced exergy analysis further elaborates on this by dividing exergy degradation into primary and secondary levels, which assists in comprehending the root causes of inefficiencies and identifying areas where genuine improvement can be achieved. The first level of exergy degradation can be split into the avoidable and unavoidable portions or the endogenous and exogenous segments.

Below, the primary differences between conventional and advanced exergy analysis will be explained, along with formulas and references. In advanced exergy analysis, exergy degradation within a component arises due to its own irreversibilities and those from other components in the system, which represents a novel perspective in exergy studies. Therefore, in this approach, the exergy degradation rate is categorized into endogenous ($\dot{E}_{D,k}^{EN}$), and exogenous ($\dot{E}_{D,k}^{EX}$) parts as follows:

$$\dot{E}_{D,k} = \dot{E}_{D,k}^{EN} + \dot{E}_{D,k}^{EX} \quad (1)$$

In another subdivision, the rate at which exergy is destroyed within a component of a system can also be categorized into avoidable ($\dot{E}_{D,k}^{AV}$) and unavoidable ($\dot{E}_{D,k}^{UN}$) components as:

$$\dot{E}_{D,k} = \dot{E}_{D,k}^{AV} + \dot{E}_{D,k}^{UN} \quad (2)$$

$$\dot{E}_{D,k,A}^{UN} = \dot{E}_{p,k,A} \left(\frac{\dot{E}_D}{\dot{E}_p} \right)_K^{UN} \quad (3)$$

Where “A” is the real operating point and $\left(\frac{\dot{E}_D}{\dot{E}_p} \right)_K^{UN}$ is the ratio at a hypostatical best efficiency. Avoidable type refers to the portion of exergy loss that can potentially be minimized or eliminated through better design, operation, or management practices. By identifying and addressing these avoidable losses, we can enhance the overall efficiency of the system and reduce waste, ultimately leading to improved performance and sustainability, whereas unavoidable type refers to the portion that cannot be diminished, regardless of the most sophisticated techniques, owing to technological restrictions [7–9].

This division of exergy destruction named as first-level or primary level exergy destruction. This classification aids in identifying and analyzing the primary mechanisms through which useful energy is dissipated, facilitating a clearer understanding of thermodynamic processes. By discerning the first level of exergy destruction, researchers and engineers can develop targeted strategies for improving energy efficiency and minimizing waste in various applications.

In addition, four new ideas are introduced by separating exergy degradation's endogenous and external parts into segments that can be avoided and those that cannot. Second-level exergy destruction is a conceptual paradigm that helps to understand the complex nature of exergy loss. Classifying these parts allows us to zero in on certain inefficiencies and fix them, leading to better exergy management and optimization tactics.

These are denoted as $\dot{E}_{D,k}^{EN,AV}$, $\dot{E}_{D,k}^{EN,UN}$, $\dot{E}_{D,k}^{EX,AV}$, and $\dot{E}_{D,k}^{EX,UN}$. These are categorized as follows:

$$\dot{E}_{D,k}^{EN} = \dot{E}_{D,k}^{EN,AV} + \dot{E}_{D,k}^{EN,UN} \quad (4)$$

$$\dot{E}_{D,k}^{EX} = \dot{E}_{D,k}^{EX,AV} + \dot{E}_{D,k}^{EX,UN} \quad (5)$$

$$\dot{E}_{D,k}^{AV} = \dot{E}_{D,k}^{EX,AV} + \dot{E}_{D,k}^{EN,AV} \quad (6)$$

$$\dot{E}_{D,k}^{UN} = \dot{E}_{D,k}^{EX,UN} + \dot{E}_{D,k}^{EN,UN} \quad (7)$$

A detailed explanation of these portions and their impact on system performance enhancement is provided below [8,23,24].

- **Endogenous Unavoidable Exergy Destruction ($\dot{E}_{D,k}^{EN,UN}$):** This part represents the portion of the exergy degradation resulting from irreversibilities inherent to the k-th component that cannot be mitigated through enhancement in that component.
- **Endogenous Avoidable Exergy Destruction ($\dot{E}_{D,k}^{EN,AV}$):** This signifies the part of the irreversibility that can be reduced by enhancing the operation of the k-th component.
- **Exogenous Unavoidable Exergy Destruction ($\dot{E}_{D,k}^{EX,UN}$):** This is the portion of irreversibility that cannot be diminished due to irreversibilities from other system components related to technological limitations.
- **Exogenous Avoidable Exergy Destruction ($\dot{E}_{D,k}^{EX,AV}$):** This represents the part of exergy degradation that can be minimized by enhancing the performance of other system components.

There are several ways to determine how much endogenous exergy degradation each part has. The thermodynamic cycle method is one such approach; it creates a hybrid thermodynamic cycle for each part so it can work in practice while the rest run smoothly in an ideal setting. This method calculates ($\dot{E}_{D,k}^{EN}$) as the quantity of hybrid cycles is associated with the number of components within the system [8,25]. This study uses the thermodynamic cycle approach due to its precision [8,23,24,26].

The exogenous exergy destruction ($\dot{E}_{D,k}^{EX}$) is computed using the values of ($\dot{E}_{D,k}^{EN}$) and the total exergy destruction ($\dot{E}_{D,k}$) from Equation (1). Equation (3) provides the value of ($\dot{E}_{D,k}^{UN}$), and using these values, ($\dot{E}_{D,k}^{AV}$) is obtained from Equation (2). The unavoidable endogenous exergy destruction ($\dot{E}_{D,k}^{EN,UN}$) is determined by operating component k under unavoidable conditions, while the others function ideally. Equations (4) and (6) are then used to derive ($\dot{E}_{D,k}^{EN,AV}$) and ($\dot{E}_{D,k}^{EX,AV}$), respectively. Lastly, ($\dot{E}_{D,k}^{EX,UN}$) is calculated using ($\dot{E}_{D,k}^{EX,AV}$) from Equation (5).

3. Evaluating conventional versus advanced exergo-economic analyses

Estimating the expenses of a system during its lifetime is an essential part of economic study. Capital expenditures, ongoing upkeep, operational expenses, and energy costs are all potential components of this total. A thermo-economic analysis is a typical tool for assessing the efficiency of energy systems. It combines economic analysis with thermodynamic principles to find the parts and procedures most concerning system efficiency and financial costs. Exergo-economic calculations are derived by integrating thermodynamic relationships with exergy analysis. By applying thermo-economic principles to the system's investment, operating, and maintenance expenses, the exergo-economic method assigns a monetary value to each exergy flow, heat transfer, and exergy destruction. The analysis's findings shed light on the system's total cost and the components' relative contributions to this total.

Advanced exergoeconomic analysis, however, enhances the conventional approach by breaking down component costs into those associated with exergy destruction, fostering more precise insights into the costs tied to exergy losses in each part of the system. This method allows for informed decision-making regarding system optimization by revealing how the exergy destruction costs of one component affect others. Unlike conventional exergoeconomic analysis, which overlooks complex component interactions, advanced exergoeconomic analysis accounts for these relationships and distinguishes costs into avoidable and unavoidable, endogenous and exogenous categories. This classification empowers engineers to target improvements where the potential for cost reduction is highest, facilitating more efficient designs. Advanced analytics yield comprehensive data, clarify the impacts of component interactions, and reveal optimization opportunities. Although it demands more intricate calculations and data, advanced exergoeconomic analysis is a more effective and diagnostic tool for enhancing energy system performance and cost allocation than conventional exergoeconomic approaches or standalone optimization methods, as it offers deeper insight into the origin and structure of system inefficiencies. Depending on the operational conditions and the interactions among components, there are different parts to the costs of exergy degradation rate and investment in advanced exergo-economic analysis. Combinations of endogenous and avoidable expenses, as well as those from standard and sophisticated exergy analyses, are used to identify the breakdown of these sectors.

Based on the above explanation, the cost of exergy destruction can be expressed as follows:

$$\dot{C}_{D,k} = \dot{C}_{D,k}^{EN} + \dot{C}_{D,k}^{EX} \quad (8)$$

$$\dot{C}_{D,k} = \dot{C}_{D,k}^{AV} + \dot{C}_{D,k}^{UN} \quad (9)$$

Similarly, the investment cost can be expressed as:

$$\dot{Z}_k = \dot{Z}_k^{EN} + \dot{Z}_k^{EX} \quad (10)$$

$$\dot{Z}_k = \dot{Z}_k^{AV} + \dot{Z}_k^{UN} \quad (11)$$

Where, $\dot{C}^{EN}$ and $\dot{C}^{EX}$ show the endogenous exergy cost rate and the exogenous exergy cost rate, respectively. Moreover, $\dot{Z}^{EN}$ and $\dot{Z}^{EX}$ are refer to the endogenous investment cost rate and the endogenous investment cost rate.

By combining the endogenous/exogenous and avoidable/unavoidable segments, four components of exergy destruction cost and investment cost can be identified. These components, along with their definitions and methods of calculation, are described below:

$$\dot{C}_{D,k}^{EN} = \dot{C}_{D,k}^{EN,AV} + \dot{C}_{D,k}^{EN,UN} \quad (12)$$

$$\dot{C}_{D,k}^{EX} = \dot{C}_{D,k}^{EX,AV} + \dot{C}_{D,k}^{EX,UN} \quad (13)$$

$$\dot{C}_{D,k}^{AV} = \dot{C}_{D,k}^{EX,AV} + \dot{C}_{D,k}^{EN,AV} \quad (14)$$

$$\dot{C}_{D,k}^{UN} = \dot{C}_{D,k}^{EX,UN} + \dot{C}_{D,k}^{EN,UN} \quad (15)$$

Similarly, the following relationships can be considered for the investment cost:

$$\dot{Z}_{D,k}^{EN} = \dot{Z}_{D,k}^{EN,AV} + \dot{Z}_{D,k}^{EN,UN} \quad (16)$$

$$\dot{Z}_{D,k}^{EX} = \dot{Z}_{D,k}^{EX,AV} + \dot{Z}_{D,k}^{EX,UN} \quad (17)$$

$$\dot{Z}_{D,k}^{AV} = \dot{Z}_{D,k}^{EX,AV} + \dot{Z}_{D,k}^{EN,AV} \quad (18)$$

$$\dot{Z}_{D,k}^{UN} = \dot{Z}_{D,k}^{EX,UN} + \dot{Z}_{D,k}^{EN,UN} \quad (19)$$

The endogenous exergy cost rate ($\dot{C}^{EN}$) and the exogenous exergy cost rate ($\dot{C}^{EX}$) can be computed as follows:

$$\dot{C}_{D,K}^{EN} = c_{F,K} \dot{E}_{D,K}^{EN} \quad (20)$$

$$\dot{C}_{D,K}^{EX} = c_{F,K} \dot{E}_{D,K}^{EX} \quad (21)$$

$$\dot{C}_{D,K} = c_{F,K} \dot{E}_{D,K} = \dot{C}_{D,K}^{EN} + \dot{C}_{D,K}^{EX} \quad (22)$$

The inescapable exergy cost rate ($\dot{C}^{UN}$) and the preventable exergy cost rate ($\dot{C}^{AV}$) are obtained as follows:

$$\dot{C}_{D,K}^{UN} = c_{F,K} \dot{E}_{D,K}^{UN} \quad (23)$$

$$\dot{C}_{D,K}^{AV} = c_{F,K} \dot{E}_{D,K}^{AV} \quad (24)$$

$$\dot{C}_{D,K} = c_{F,K} \dot{E}_{D,K} = \dot{C}_{D,K}^{UN} + \dot{C}_{D,K}^{AV} \quad (25)$$

The inescapable investment cost rate ($\dot{Z}_K^{UN}$) is derived from the block's inefficiency and is computed using Equation (26).

$$\dot{Z}_K^{AV} = \dot{Z}_K - \dot{Z}_K^{UN} \quad (26)$$

The endogenous exergy cost rate ($\dot{C}^{EN}$), exogenous exergy cost rate ($\dot{C}^{EX}$), unavoidable exergy cost rate ($\dot{C}^{UN}$), and avoidable exergy cost rate ($\dot{C}^{AV}$) are elaborated upon in further detail.

The unavoidable endogenous exergy cost rate ($\dot{C}_D^{UN,EN}$) and the

inescapable endogenous investment cost rate ($\dot{Z}_K^{UN,EN}$) cannot be reduced by altering "the technical aspects and economic circumstances of the specific component. The calculating formulas are presented in Equations (27) and (28).

$$\dot{C}_{D,K}^{UN,EN} = c_{F,k} \dot{E}_{D,K}^{UN,EN} \quad (27)$$

$$\dot{Z}_K^{UN,EN} = \dot{E}_{P,K}^{UN,EN} \left(\frac{\dot{Z}_K}{\dot{E}_{P,K}} \right)^{UN} \quad (28)$$

The calculation formula is given in Equations (29) and (30).

$$\dot{C}_{D,K}^{UN,EX} = c_{F,k} \dot{E}_{D,K}^{UN,EX} \quad (29)$$

$$\dot{Z}_K^{UN,EX} = \dot{Z}_K^{UN} - \dot{Z}_K^{UN,EN} \quad (30)$$

The avoidable endogenous exergy cost rate destruction ($\dot{C}_D^{AV,EN}$) and the unavoidable endogenous investment cost rate ($\dot{Z}_K^{AV,EN}$) can be declined by enhancing the block itself. The formula for calculation is presented in Eq. (31) and Eq. (32).

$$\dot{C}_D^{AV,EN} = c_{F,k} \dot{E}_{D,K}^{AV,EN} \quad (31)$$

$$\dot{Z}_K^{AV,EN} = \dot{Z}_K^{EN} - \dot{Z}_K^{UN,EN} \quad (32)$$

The avoidable exogenous exergy cost rate destruction ($\dot{C}_D^{AV,EX}$) and the avoidable exogenous investment cost rate ($\dot{Z}_K^{AV,EX}$) can be reduced by boosting other components. The calculation formulas are provided in Eq. (33) and Eq. (34).

$$\dot{C}_{D,K}^{AV,EX} = c_{F,k} \dot{E}_{D,K}^{AV,EX} \quad (33)$$

$$\dot{Z}_K^{AV,EX} = \dot{Z}_K^{EX} - \dot{Z}_K^{UN,EX} \quad (34)$$

The relationship between product exergy and feed exergy is referred to as exergy efficiency, as illustrated in the formula provided below [20]:

$$\varepsilon_K = \frac{\dot{E}_{P,K}}{\dot{E}_{F,K}} = 1 - \frac{\dot{E}_{D,K}}{\dot{E}_{F,K}} \quad (35)$$

$$\varepsilon_{K,modified} = \frac{\dot{E}_{P,K}}{\dot{E}_{F,K} - \dot{E}_{D,K}^{UN} - \dot{E}_{D,K}^{AV,EX}} \quad (36)$$

where ε represents the exergy efficiency.

Equations (37) and (38) can be used to get the exergy destruction ratio (y).

$$y_{D,K} = \frac{\dot{E}_{D,K}}{\dot{E}_{D,tot}} \quad (37)$$

$$y_{D,K}^{adv} = \frac{\dot{E}_{D,K}^{EN,AV}}{\dot{E}_{D,tot}} \quad (38)$$

Eqs. (39) and (40) are employed to calculate the exergo-economic factor (f), which denotes the impact of thermodynamic and economic characteristics on elements.

$$f_K = \frac{\dot{Z}_K}{\dot{Z}_K + \dot{C}_{D,K}} \quad (39)$$

$$f_K^{AV,EN} = \frac{\dot{Z}_K^{AV,EN}}{\dot{Z}_K^{AV,EN} + \dot{C}_{D,K}^{AV,EN}} \quad (40)$$

The total cost rate can be determined by employing Eq. (41) and Eq. (42).

$$\dot{R}_{tot} = \dot{Z}_K + \dot{C}_{D,K} \quad (41)$$

$$\dot{R}_{tot}^{AV,EN} = \dot{Z}_K^{AV,EN} + \dot{C}_{D,K}^{AV,EN} \quad (42)$$

4. Evaluating conventional versus advanced exergo-environmental analysis

Conventional or traditional exergo-environmental analysis focuses on evaluating the environmental impacts related to exergy destruction within a system. This method assesses the system's components and processes to estimate their contributions to overall environmental degradation. It integrates thermodynamic principles with environmental metrics, providing insights into the environmental cost of irreversibilities. However, it simplifies interactions by considering the system's components independently and assumes average values for the environmental impacts, which can lead to generalized conclusions. Advanced Exergo-Environmental Analysis, on the other hand, provides a more detailed and precise evaluation by decomposing the environmental impacts into endogenous and exogenous component as well as avoidable and unavoidable components. Endogenous impacts arise from internal irreversibilities of a component, while exogenous impacts are due to the interactions between components. Similarly, design or operational improvements can mitigate avoidable impacts, while technological or economic limitations constrain unavoidable impacts. This decomposition enables a deeper understanding of how specific components and their interactions affect the system's environmental performance. By comparing these approaches, it becomes evident that the advanced method is superior in identifying targeted improvement opportunities. While the conventional approach offers a broader view, advanced exergo-environmental analysis highlights the specific areas with the highest potential for environmental impact reduction. However, this comes at the cost of increased computational complexity and the need for more detailed data. Despite this, the advanced method provides a more comprehensive framework for designing environmentally sustainable energy systems. The primary objective of exergo-environmental analysis is to pinpoint the elements that have the significant influence on the environment throughout the system's life cycle, as well as those that contribute substantially to environmental pollution due to exergy degradation. So, it can assess elements that produce pollutant emissions [27]. The initial part of the method is evaluating the environmental effects integrated include all system components in the life cycle assessment (LCA) method. The rate of environmental impacts during the three phases of a component's lifecycle: construction, operation and maintenance, and disposal is determined through LCA [28]. Every component of the system generates specific environmental impacts (EIs) throughout various stages of its life cycle. It is essential to evaluate all phases of the component life cycle to accurately assess the environmental impacts, utilizing LCA. While various methodologies exist for conducting life cycle assessment (LCA), including: ReCiPe 2016, CML, and IPCC approaches, exergoenvironmental analysis, as an integrated approach combining exergy and LCA, applies a specific framework for the allocation of environmental impacts based on exergy flows and destruction. This approach employs the following balance equations [29]:

$$\dot{B}_{in,k} + \dot{Y}_k + \dot{B}_k^{PF} = \dot{B}_{out,k} \quad (43)$$

$$\dot{B}_k = \dot{E}_k \cdot b_k \quad (44)$$

$$\dot{Y}_k = \dot{Y}_k^{CO} + \dot{Y}_k^{OM} + \dot{Y}_k^{DI} \quad (45)$$

$$f_{b,k} = \frac{\dot{Y}_k}{\dot{Y}_k + \dot{B}_{D,K}} \quad (46)$$

In this context, $\dot{B}_{in,k}$ and $\dot{B}_{out,k}$ represent the rates of environmental

impact associated with the input and output exergy streams of component k , respectively. The term $\dot{Y}_k$ denotes the environmental impact specific to the component itself. Additionally, $\dot{B}_k^{PF}$ indicates the environmental impact rate attributed to emissions, while b_k refers to the specific environmental impact rate. The environmental impact rates arising during the development, operation, maintenance, and disposal phases of the component are represented by $\dot{Y}_k^{CO}$, $\dot{Y}_k^{OM}$ and $\dot{Y}_k^{DI}$, respectively. The exergo-environmental factor is denoted by $f_{b,k}$, and $\dot{B}_{D,k}$ signifies the environmental impact rate due to exergy destruction.

Conventional exergo-environmental analysis identifies the components that require to be enhanced, yet it doesn't define real degree to which these components can be optimized. Advanced Exergo-environmental analysis closely resembles advanced exergy analysis, as it is conducted based on conventional exergo-environmental analysis. The environmental effect rate of exergy destruction is categorized as endogenous/exogenous and avoidable/unavoidable factors. Advanced exergo-environmental analysis could reduce the significant emission resulting from the attempt for optimal performance. The analysis results indicate that system efficiency and environmental protection can be more effectively matched [30]. The primary focus is on the avoidable endogenous exergy destruction rate and the exergy destruction environmental effect rate. They indicate the capacity to decrease the rates of exergy destruction, exergy destruction costs, and exergy destruction environmental impacts through the modification of technological limits and the enhancement of component efficiency. By computing the average environmental impact per exergy unit of the fuel ($b_{F,K} = \dot{B}_{F,K} / \dot{E}_{F,K}$) [31]. The advanced exergy destruction environmental impact rates for each component are presented in Table 1, according to the following relationships [30]:

$$\dot{B}_{D,k} = \dot{B}_{D,k}^{EN} + \dot{B}_{D,k}^{EX} \quad (47)$$

$$\dot{B}_{D,k} = \dot{B}_{D,k}^{UN} + \dot{B}_{D,k}^{AV} \quad (48)$$

$$\dot{B}_{D,k} = \dot{B}_{D,k}^{UN,EN} + \dot{B}_{D,k}^{AV,EN} + \dot{B}_{D,k}^{UN,EX} + \dot{B}_{D,k}^{AV,EX} \quad (49)$$

Diagram of equations related to advanced exergy analysis, advanced exergo-environmental analysis, and also advanced exergo-economic analysis are presented in Fig. 1.

5. Different methods of advanced exergy analysis

There are several ways to break down exergy destruction into its endogenous, external, preventable, and inescapable parts. These methods vary regarding their assumptions, intricacy, data prerequisites, and relevance to actual systems. The thermodynamic cycle approach, engineering (graphical) method, exergy balancing method, equivalent component method, structural theory method, and the modified hybrid method are the most popular ones [4,5,8,11,34–39]. To figure out which of these strategies is best for different sorts of energy systems, it's

important to compare them and understand how they work. Table 2 shows how various strategies are similar and different in a qualitative way. Although not incorporated in this study, the application of these methodologies to a unified case study (e.g., a gas turbine cycle) would provide a more stringent comparative framework.

5.1. Thermodynamic cycle method

The thermodynamic cycle method is one of the most systematic and reliable ways to get measurable information on all parts of exergy destruction. It looks at exergy flows and destructions to figure out how well whole thermodynamic cycles (such Rankine, Brayton, and refrigeration) are working and how to make them work better. This method focuses on studying how exergy is spread out and broken down over full cycles instead of only looking at individual parts. There are four reference cycles: the actual thermodynamic cycle, the unavoidable thermodynamic cycle, hybrid thermodynamic cycle I, and hybrid thermodynamic cycle II. These cycles help separate the impacts of interactions between components and their own constraints. Using them makes it feasible to measure the exergy loss for each part, which helps to fully understand how the system works and where it may be improved [34,36,39].

5.2. Engineering (graphical) method

This method is more practical and application-focused, emphasizing real-world system design, operation, and optimization. It incorporates exergy analysis into standard engineering evaluation practices and visualizes endogenous and exogenous components through sensitivity analysis and graphical tools. It is particularly applicable to systems with chemical reactions. However, it may fail to determine endogenous destruction if a specific base condition is not met [8,9].

5.3. Exergy balance method

This method is based on the first and second laws of thermodynamics and quantifies exergy flows into and out of each system component to determine irreversibilities. It is effective in identifying inefficiencies at the component level and has been applied in gas turbines, compressors, and heat exchangers. It does not account for component interactions explicitly but remains a foundational tool in exergy studies. Kelly demonstrated its application to gas turbine combustion chambers [8,9].

5.4. Equivalent component method

This method simplifies complex systems by replacing real components with idealized equivalents. It helps evaluate how much a real component deviates from its ideal model and quantifies the resulting exergy destruction. Although useful in identifying inefficiencies and suggesting component-level improvements, its reliance on different working fluids between real and idealized states is a limitation [8,9].

5.5. Structural theory method

Developed by Valero and colleagues, this method evaluates how a component's malfunction or dysfunction affects the rest of the system. It distinguishes between additional fuel consumption due to local irreversibilities and those arising from system-wide interactions. Unlike other methods, it introduces the concepts of malfunction and dysfunction rather than just endogenous and exogenous splits. Structural theory has been successfully applied in real-world systems, including power plants [11].

5.6. Modified hybrid method

Proposed by Fallah et al. [35], this approach integrates elements of

Table 1

Advanced exergy destruction cost rate and environmental impact rate [32,33].

Type	Exergy destruction environmental impact rate
Endogenous	$\dot{B}_{D,k}^{EN} = b_{F,K} \cdot \dot{E}_{D,k}^{EN}$
Exogenous	$\dot{B}_{D,k}^{EX} = b_{F,K} \cdot \dot{E}_{D,k}^{EX}$
Avoidable	$\dot{B}_{D,k}^{AV} = b_{F,K} \cdot \dot{E}_{D,k}^{AV}$
Unavoidable	$\dot{B}_{D,k}^{UN} = b_{F,K} \cdot \dot{E}_{D,k}^{UN}$
Avoidable endogenous	$\dot{B}_{D,k}^{AV,EN} = b_{F,K} \cdot \dot{E}_{D,k}^{AV,EN}$
Avoidable exogenous	$\dot{B}_{D,k}^{AV,EX} = b_{F,K} \cdot \dot{E}_{D,k}^{AV,EX}$
Unavoidable endogenous	$\dot{B}_{D,k}^{UN,EN} = b_{F,K} \cdot \dot{E}_{D,k}^{UN,EN}$
Unavoidable exogenous	$\dot{B}_{D,k}^{UN,EX} = b_{F,K} \cdot \dot{E}_{D,k}^{UN,EX}$

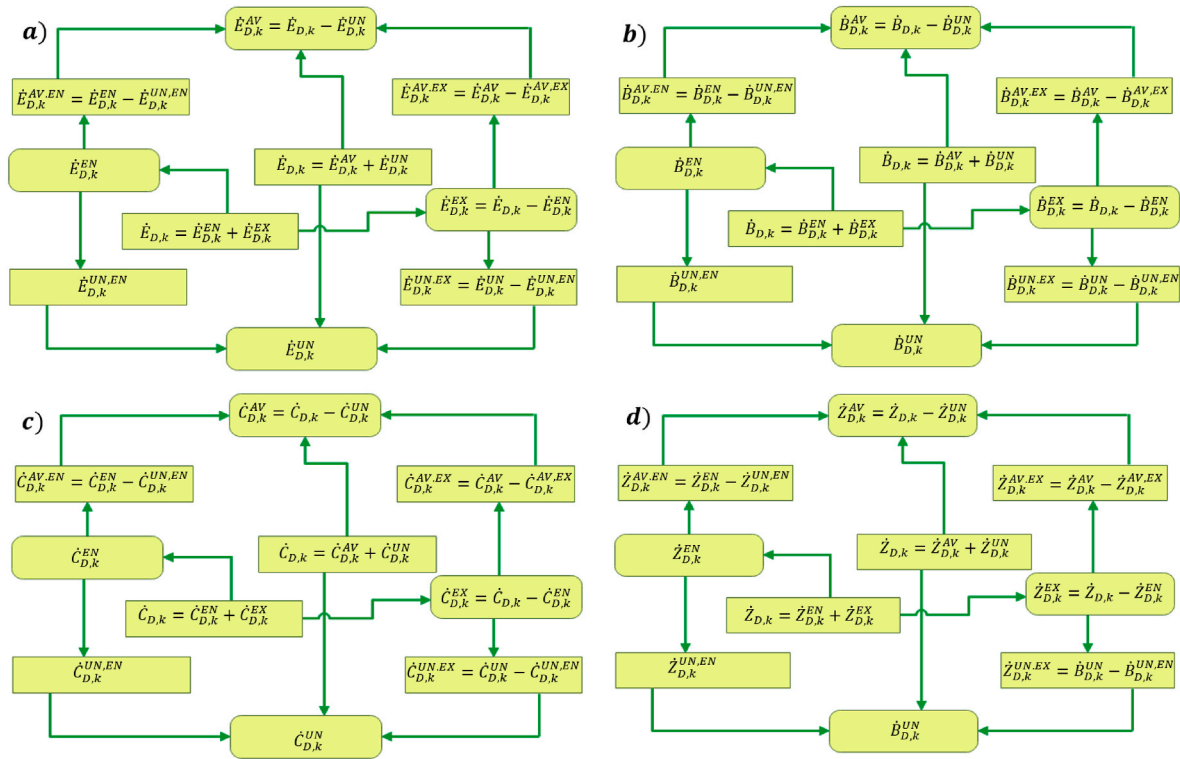


Fig. 1. Equation diagram of a) Advanced exergy analysis, b) Advanced exergo-environmental analysis, c, d) Advanced exergo-economic analysis.

Table 2

Comparative overview of advanced exergy analysis methods.

Method	Accuracy	Complexity	Applicability	Advantages	Disadvantages	References
Engineering (Graphical)	Moderate	Low	High	- Accurate for simple/complex systems. - Directly applicable to real systems.	- Cannot analyze dissipative devices (e.g., throttle valves). - Time-consuming due to iterative simulations. - Computationally intensive.	[8,9,34,36,39]
Thermodynamic Cycle	High	High	Moderate	- Theoretically universal. - Rigorous theoretical foundation.	Ideal reactor definition challenging (e.g., combustion chambers).	[8,9]
Exergy Balance	Moderate	Low	High	- Fast determination of endogenous values. - Simple implementation.	- Ignores component interactions. - Assumptions reduce accuracy (though errors <3 % vs. Engineering method).	[8,9]
Equivalent Component	Moderate	Moderate	High	Approximates complex components (e.g., reactors as heat exchangers).	- Mismatch between ideal/real fluid behavior. - Limited validation in large systems.	[4,8,9]
Symbolic Mathematics	Low	Moderate	Low	- Rapid algebraic solutions. - Useful for theoretical studies.	- Neglects mass flow dynamics. - High errors (>20 %) in complex systems	[9]
Structural Theory	High	High	High	- Captures fuel-product relationships. - Precise for system interactions. - Validated for real plants	- May require detailed modeling of subsystem interactions in large-scale applications. - Setup process can be complex for multi-product configurations.	[9,11]
Modified Hybrid	High	Moderate	High	- Optimized Engineering method - Mass/exergy conservation maintained - Superior for parametric optimization	Requires reversible Heat exchanger modeling	[35]

both the hybrid cycle and engineering methods. It employs hypothetical reversible elements to estimate minimum destruction levels and is particularly suited for fuel cell and combustion-based systems. It balances accuracy with reduced computational time. Also, modified hybrid method achieves high accuracy while resolving mass/exergy balance violations of conventional hybrid approaches.

While multiple advanced exergy analysis techniques exist, their applicability varies significantly based on system complexity, data requirements, and analytical objectives. Table 2 evaluates six established methodologies against five key criteria: accuracy, computational complexity, applicability scope, advantages, and disadvantage. The comparison framework draws on foundational work by Tsatsaronis et al. [3], Kelly et al. [8,9], Valero et al. [38], and Fallah et al. [35]. In systems

with chemical reactions, the engineering method provides the highest accuracy but can be time-consuming, especially for parametric analyses. Conversely, the improved hybrid method reduces these time constraints while maintaining accuracy and addressing issues related to mass and energy balances. The thermodynamic cycle approach demonstrates particular strength in systems with chemical reactions (e.g., combustion chambers), where its simplified idealization outperforms structural theory in computational efficiency while maintaining accuracy within 1–3 % of engineering methods. This approach is especially advantageous in parametric analyses, offering lower computational time and higher efficiency without significantly sacrificing precision.

6. Case studies

Morosuk and Tsatsaronis [7] conducted a comprehensive exergy analysis of an absorption refrigeration machine (ARM). The results demonstrated that while the conventional exergy analysis suggests the precedence of improving parts: absorber, generator, condenser and evaporator, but based on outcomes of advanced exergy analysis, the generator and then the absorber are the priority for improvement. Also, Morosuk and Tsatsaronis [40] evaluated refrigeration systems utilizing various working fluids concerning advanced exergy. The findings showed that, in general, all exergy destruction values rely on the working fluid. Also, with the goal of improving the thermodynamic efficiency of vapor-compression refrigeration systems, the priority is to improve the evaporator and then the compressor. In 2009, Morosuk and Tsatsaronis [41] reviewed advanced exergy analysis for systems involving chemical reactions and used it to analyze a basic open gas-turbine system was reviewed. Since more than 70 % of the total exergy destruction within the whole system was unavoidable, the capacity for enhancement the efficiency of the whole system was estimated to be 5 %. It was examined a combined cycle power facility utilizing conventional and advanced exergy analysis. Petrakopoulou et al. [17] according to the results of advanced exergy analysis, the improvement priority was considered for combustion chamber, expander and compressor respectively. Advanced exergy analysis for the configuration of a combined power plant with biomass gasification, in order to determine the opportunity for enhancement the system by the system by taking consideration the mutual effect of components on each other, was carried out by Soltani et al. [42]. In this research, it was found that the components impact on each other is not so strong and the endogenous exergy destruction of each part is more than the exogenous exergy destruction. Also, the advanced exergy analysis in contrast to the conventional exergy analysis showed the enhancing of the cycle efficiency, it should not be focused on the gasifier or the combustion chamber unit, but on the heat exchanger. Fallah et al. [26] compared advanced and traditional exergy investigation of the low-grade geothermal source Kalina cycle. This comparison suggests that advanced exergy analysis can help identify more nuanced opportunities for enhancing system efficiency compared to traditional approaches. The advanced exergy study determined that the condenser, turbine, and evaporator are prioritized for upgrading. However, in traditional exergy calculations, the exergy destruction rate of the evaporator is greater than that of the turbine. Fallah et al. [43] presented a comparative analysis of diverse gas turbine cycles through the utilization of advanced exergy analysis techniques. The authors assess the performance of various configurations, determining the most efficient cycle based on exergy efficiency and destruction rates. Notably, the findings reveal that the combustion chamber is the predominant exergy destruction source in the analyzed cycles, indicating that enhancements in this component may yield substantial efficiency improvements. The paper underscores the critical role of advanced exergy analysis in optimizing energy systems by effectively identifying specific components where irreversibilities can be minimized. They also conducted a thorough investigation into anode gas recirculation in solid oxide fuel cells (SOFCs) by employing advanced exergy analysis to evaluate performance and pinpoint inefficiencies. Their detailed examination of system components disclosed significant exergy losses attributable to irreversible processes during operation. This study seeks to optimize the operational parameters of SOFCs by mitigating these losses, ultimately enhancing overall efficiency. Results suggest that targeted design and operational modifications could substantially improve the energy output and sustainability of SOFC systems [44]. Ghiasi et al. [45] introduced an innovative optimization method for combined cycle systems through a detailed examination of exergy destruction at the primary level. Their research elucidates the contributions of various components to overall system inefficiencies and proposes strategies for reallocating exergy losses to enhance performance. The study illustrates that a comprehensive understanding of

exergy distribution can facilitate the development of more effective methods to enhancement both the efficiency and sustainability of combined cycle systems. The findings highlight the substantial potential for advancements in energy systems achieved through meticulous exergy analysis.

Considering both the quality and quantity of energy, through exergy analysis serves as a valuable theoretical method for enhancing efficiency of energy conversion in systems. Energy analysis is frequently employed to demonstrate the critical parameters influencing system performance. The conventional 3E analyses continue to reveal specific limits. They provide an overall indication of the energy lost during component operation, without specifying the potential for reduction in energy loss, consequently. Many researchers have enhanced conventional 3E (Energy, Exergy, and Economic (studies and introduced advanced exergy analysis, advanced exergoeconomic analysis, and advanced exergoenvironmental analysis. Advanced 3E analyses recognize between exergy destruction, the cost of exergy destruction, and the environmental impact of exergy destruction effect based on endogenous/exogenous and avoidable/unavoidable classifications, facilitating the assessment of components' potential for improvement through the optimization of their efficiency and technical conditions. The study conducted a conventional and advanced exergy analysis of the recompression supercritical carbon dioxide cycle, indicating that to enhance system efficacy, focus should be assigned first to the reactor, followed by the pre-cooler, low-temperature heat exchanger, high-temperature heat exchanger, and the turbine. However, the advanced exergy analysis focused on the following components: the high-temperature recuperator, the turbine, and the main compressor [29]. A technique for recovering waste heat utilizing a recuperative organic Rankine cycle (RORC) regenerator, integrated with the exhaust gas pipeline of a gas-powered engine, was assessed from both environmental and thermodynamic perspectives. The investigation indicated that a significant portion of the exergy destruction was endogenous, amounting to 105.08 kW (81.6 %). The environmental effects related to the turbine for aluminum were quantified as 27,617.21 kg of CO₂ equivalent [46]. In the investigation of a cascade absorption heat transformer was suggested for utilizing industrial low-grade waste heat. The advanced exergoeconomic parameter revealed the optimizing priorities for the LiBr absorber, NH₃ condenser, and NH₃ absorber differ from that suggested through conventional approaches [47]. In the thermoeconomic system analysis, the unit cost of the product, investment cost, and the exergy destruction cost for the system components are computed. The thermoeconomic assessment of the superconducting carbon dioxide cycle suggests abandoning enhancements to the turbine and reactor due to their significant economic cost, while focusing on the optimization of other cycle components by advanced exergy analysis prioritizing [48]. For the initial time, the thermodynamic efficiency of significant transcritical R744 supermarket refrigeration system structures has been compared using advanced exergy analysis [49]. Pump thermal electricity storage (PTES) represents a thermo-mechanical energy storage system. This study conducts both Conventional and advanced exergy evaluations of a 10 MW JB-PTES, attempting to fulfill a gap in the existing literature through the identification of the components for the greatest irreversibility [50]. The utilization of hot dry rocks as a substitute for fossil fuels could considerably reduce environmental pollution. This research examines the impact of initial operating parameters on the thermodynamic efficiency of a trans-critical CO₂ cycle, which includes a pressurization mechanism for utilizing hot dry rock resources [51]. The potential for improvement and the limitations of the SOFC system integrated using a kinetic-based modeling approach for the pre-reformer were examined. Conventional exergy study indicates that enhancing the Stack and Heat exchanger is crucial, contributing 32.54 % and 30.19 % of the total exergy destruction, respectively. The advanced exergy study indicates that importance for improvement should be assigned to the Stack and Afterburner [52]. A novel liquid carbon dioxide energy storage system containing a cold recuperator and

low-pressure storage has been introduced. Both conventional and advanced exergy analyses suggest that optimizing the low-pressure compressor should be prioritized [53]. The optimization of the recuperative organic Rankine cycle (RORC), including a gas turbine unit and spray dryers in a ceramic industry, was conducted under some distinct situations. Advanced exergy study indicates that the combustion chamber (CC) is not just one component responsible for exergy destruction; chemical processes that occur in the CC, wall tile dryer (WD), and ground tile dryer (GD) contribute to an enhanced endogenous exergy destruction rate within the COGEN system [13]. A conventional coal-to-SNG system was examined, revealing that the avoidable exergy destruction constitutes 61.88 % of the total exergy destruction, indicating substantial potential for enhancement in this coal-to-synthetic natural gas (SNG) production process [54]. The advanced exergy analysis, along with the conventional energy analysis, was utilized to develop and enhance the CO₂ liquefaction system employing a heat pump. The total exergy destruction for the heat pump system utilizing R290 and R717 is measured at 2660.73 kW and 2457.26 kW [55]. Combi boilers are frequently utilized for heating in houses, particularly since the increasing use of natural gas as a fuel source. An advanced exergy analysis was performed to analyze the capacity for improvement in combi boilers by identifying both ideal and unavoidable conditions, including the combustion process [56]. In contrast to the high-temperature compressed carbon dioxide energy storage (CCES) system, the CCES system operating under low-temperature conditions shows greater technological advantages. Both conventional exergy analysis and advanced exergy analysis were employed to examine the thermodynamic properties of a low-temperature compressed carbon dioxide energy storage (LT-CCES) system, which includes a CO₂ Brayton cycle, low-temperature thermal energy storage, and cold energy storage, with the thermal energy storage temperature maintained below 200 °C [57]. The working fluid selection and advanced exergy analysis for the dual-loop organic Rankine cycle (DORC) system driven by flue gas at 300 °C were conducted. The objective functions chosen were exergy efficiency, payback period, and annual CO₂ emission reduction. The endogenous avoidable portion of the DORC system is estimated to be around 53.5 % [58]. An advanced exergetic method was utilized to evaluate the exergetic efficiency within a CO₂ trans-critical ejector refrigeration system that includes a thermoelectric sub-cooler. Improvements to the compressor and thermoelectric sub-cooler should be prioritized according to both conventional and advanced exergy approaches [59]. The Modified Productive Structure Analysis (MOPSA) method, which provides a more precise and thorough depiction of exergy flows and their related expenses, was introduced as part of the advanced exergo-economic investigation of an advanced adiabatic compressed air energy storage system (AA-CAES) that utilized both conventional and cutting-edge methodologies. By using this cutting-edge technique, the scientists were able to isolate the endogenous and external factors contributing to exergy degradation, shedding light on where the system may use improvement. The results demonstrated that modifying critical system components like compressors and turbines could significantly improve economic performance by lowering exergy destruction. The AA-CAES system could achieve significant efficiency and cost reductions through the combination of sophisticated analysis and multi-objective optimization using NSGA-II. This approach is precious for future energy storage applications [47]. The advanced exergy analysis, which categorizes irreversibility into endogenous/exogenous and avoidable/unavoidable segments, illustrates the increaseable irreversibility component inside every cycle component. In the hybrid S-CO₂/ORC system, results indicated that 34.76 % of the total exergy destruction was avoidable and could be minimized. Furthermore, 61.40 % of the avoidable portion was endogenous, which can be reduced through by modifying the components themselves [24]. An analysis of both traditional and advanced exergy investigations for power and cooling cogeneration system based on an ammonia-water cycle, connected using a low-temperature geothermal heat source for distributed

cooling applications, was provided. The thermodynamic cycle approach is employed for advanced exergy analyses [34]. An advanced exergy study of the combination cycle including a helium reactor gas turbine coupled with an organic Rankine cycle was conducted utilizing equation-solving engineering software. The thermodynamic method for modeling of this cycle was performed, and the energy and traditional exergy analyses the results were obtained [26]. The research evaluated the optimal working fluid and operating conditions for a solid waste-powered ORC-CHP system through traditional and advanced exergy analysis. The rate of exergy degradation of the endogenous components exceeds that of the exogenous components, with 99.44 % of the total exergy destruction attributed to the internal irreversibility of the system's components [60]. A novel integrated process utilizing torrefaction, gasification, and solid oxide fuel cells (SOFC) has been established for the valorization of biomass waste. An investigation of process sustainability has been conducted using economic, exergy, advanced exergy, and exergoeconomic indicators [61]. The ejector is significant for enhancing the CO₂ trans-critical refrigeration system; therefore, the exergetic efficiency of a CO₂ two-phase ejector was thoroughly examined utilizing advanced exergy analyses techniques. The interconnections among the ejector parts are weak, and the endogenous irreversibility of the ejector is significant [62]. Advanced exergy analysis is conducted to evaluate a suggested combined system consisting an ORC, a proton exchange membrane fuel cell (PEMFC), and a hydrogen storage unit. The analysis was developed using the Intel Fortran compiler integrated with Microsoft Visual Studio [63]. Research indicates that, advanced exergo-economic analysis has been implemented in an integrated power production cycle utilizing the cold energy of LNG through the MOPSA approach. 23 % of the exergy degradation and the associated cost rate of the integrated system was avoidable [64]. Advanced exergetic, exergoeconomic, and exergoenvironmental analyses of a comprehensive olefin plant were performed for the first time, with environmental effects calculated through LCA [31]. The exergo-economic analysis of an integrated solar combined cycle (ISCC) system studies compressors, turbines, and heat recovery steam generators (HRSGs) for exergy destruction and efficiency. Advanced exergy methods like phase change material thermal energy storage improve energy and economic performance. This innovative method differentiates preventable and unavoidable exergy losses and internal/external interactions, providing deeper system optimization insights than older methods. Molten salt outperformed water and Thermion oil in heat transfer, while solar integration and heat storage increased system efficiency and lowered CO₂ emissions. The analysis also shows that sophisticated methodologies can improve ISCC system design and sustainability by allocating money and optimizing components with the highest exergy degradation [65]. Gürbüz et al. [66] performed an exergo-environmental analysis of an energy system integrated with geothermal energy sources and a two-stage ORC. The environmental effects were categorized into endogenous and exogenous, as well as avoidable and unavoidable portions. This classification enabled a deeper understanding of how different components interact with one another and highlighted the opportunities for optimization. Analyzing these impacts through an environmental lens allowed for a more comprehensive evaluation of the system's efficiency and the potential benefits of reducing detrimental effects on the environment, as discussed by Nourpour et al. [67]. They conducted a detailed investigation of a natural gas injection facility, employing advanced exergy-based, exergo-economic, and exergo-environmental assessment through the use of gas turbines to enhance the comprehension of the energy system. The findings demonstrated that, relative to the objectives of the employer, the system's efficiency and its components may be enhanced through advanced analysis. The findings indicated that the absorber unit exhibited the highest levels of unavoidable and endogenous exergy destruction, while the combustion chamber was responsible for the most significant unavoidable environmental effect related to exergy destruction. The review of literature illustrates that advanced

exergy-based assessment of energy conversion systems may reveal actual potential for minimizing thermodynamic irreversibility and enhancing environmental efficacy in energy production processes. The impact of various configurations on the exogenous exergy degradation of each component for S-CO₂ was investigated. It revealed the inter-cooling cycle exhibits the minimal exergy destruction value relative to alternative configurations, indicating that the pre-compression cycle presents the most potential for improvement [68]. To assess the accurate potential for enhancing an innovative integrated cooling, heating, and power production system supplied by geothermal energy by geothermal energy, the first and second levels of exergy degradation were evaluated. The overall rate of preventable exergy loss, representing the system's maximum enhancement potential is 9732.4 kW, with 60.01 % of this avoidable value is endogenous, whereas 39.99 % is exogenous [69]. An advanced exergy analysis was conducted on the oil and gas processing plant located on a North Sea platform for a specific production day. This analysis indicated that the inefficiencies of the compressors stem from their intrinsic irreversibility, whereas the exergy destruction in the coolers could be specifically reduced by enhancing the other components of the system [70]. A recuperative ORC was developed using a three-fluid condenser for the utilization of cryogenic cold energy recovery. The conventional exergy study revealed the condenser as the primary focus for system optimization, responsible for 62.6 % of the overall exergy destruction. The rate of endogenous exergy that could have been avoided in the operation of the expander was calculated to be 54.0 %. This indicates that a significant portion of exergy loss, specifically 54.0 %, could have been prevented through optimal design or operational strategies [71]. The poor heat-to-power conversion performance of geothermal sources restricts their growth and consumption; a comprehensive investigation of the factors contributing to irreversible losses and the costs associated with each component of the geothermal power generation system could present theoretical assistance for enhancing its efficacy. An investigation and evaluation of an organic Rankine flash cycle (ORFC) power production system is analyzed. The avoidable endogenous exergy degradation/cost is 54.07 % and 48.94 % of the overall exergy degradation/cost of the system [72]. The components of the evaporator and domestic water heater (DWH) must be prioritized to enhance the performance of cogeneration cooling, heating, and power (CCHP) system by using the ejector-based CO₂ trans-critical cycle. The results of exogenous exergy destruction indicate that the condenser's exergy destruction was -0.468, suggesting that the exergy destruction of the DWH can be minimized [73]. The exergy performance of a solar-powered cascade ejector system with thermal storage was analyzed, involving a solar-driven ejector heat storage segment and an ejector-based CO₂ trans-critical refrigeration segment. The enhanced exergy assessment of the system indicates that the R290 ejector in the solar-powered ejector heat storage subsystem (SEHSS) provides the greatest potential for enhancement [74]. A multi-effect desalination system, integrated with a sugar industry unit and utilizing the ORC cycle for power and freshwater production, has been designed. An advanced exergy analysis is performed on every component of the system. under its operating conditions to determine the contribution of avoidable irreversibilities [75]. The non-dominated sorting genetic algorithm (NSGA-II) was applied for the poly-criteria optimization of AA-CAES. The total unit cost of the product and exergy efficiency for the conventional and advanced exergy/exergoeconomic the analyses determined the value to be 162.9 \$/day. and 64.3 %, and 103.5 \$/day and 91.9 %, respectively [65]. Furthermore, a novel renewable energy driven wind-solar-hydrogen multi-energy supply (WSH-MES) integrated electricity, hydrogen, heating and storage system was presented. A dual-level capacity-operation co-optimization system was significantly proposed for the WSHMES system to enhance power generation and system performance [76]. The coal to ethylene glycol (CtEG) model suffers challenges due to its significant energy consumption and inefficient thermodynamic performance. The CtEG process was both presented and modeled using Aspen Plus software,

resulting in the obtaining of the material and energy balance for all stages of the process [77]. A combined power and cooling system with the recovery of waste heat, adaptable cooling, and hydrogen energy usage is employed in data centers for achieving carbon peak and carbon neutrality; performance modeling and advanced exergy analysis of the system were performed [78]. An advanced exergo-economic analysis should include an energy-exergy-economic-environmental assessment, sustainability, and sensitivity methodologies to assess single-slope solar still's efficiency, cost-effectiveness, and environmental impact to reduce water scarcity. The authors explain how to reduce system costs and improve performance by integrating new exergo-economic and energy-environmental methods, making water desalination systems more sustainable and efficient. Advanced exergo-economic analysis, especially comparing solar desalination systems to alternative energy systems, benefits from this [78]. Desalinates are evaluated using advanced exergo-economic analysis. Bagasse and MSW power the organic Rankine cycle and MED plant. Avoidable and unavoidable system component exergy degradation is determined by advanced exergy analysis. 15 % of system exergy loss is preventable, and 85 % is inevitable due to technology restrictions. Component inefficiencies contribute 76 % of exergy degradation, while component interactions cause 24 %. Exergo-economic analysis assigns system costs by exergy destruction and capital spending. Results demonstrate turbines and compressors improve thermodynamic and economic efficiency. This comprehensive approach shows that better exergy and exergo-economic evaluations help optimize energy systems [79]. The research conducted an advanced exergy investigation of the ejector cascade refrigeration cycle. The investigation revealed that the avoidable exergy destruction in the system constitutes 38.674 %, whereas the endogenous and exogenous irreversibility represent 77.772 % and 22.228 % of the overall exergy degradation, respectively [80]. The suggested systems include SOFC, GT, HRSG, Rankine cycle, and heat exchanger. In contrast to conventional analyses and advanced analyses reveal that air compressors contain the highest avoidable-endogenous cost rates [81]. Moreover, solar dryers significantly contribute to the reducing of greenhouse gas emissions, and conventional and enhanced exergy assessment of an innovative indirect solar dryers have been examined. A trans-critical CO₂ storage system was developed, combining heat source and solar energy to improvement the deployment of renewable energy in energy-related applications. The maximum avoidable exergy loss of the system is 4.17 MW, accounting for 68.77 % of the overall exergy degradation [82]. The air-source auto-cascade high-temperature heat pump (AHTHP) can employ air to generate high-temperature steam, beneficial for decarbonization of industry. An air source AHTHP was simulated utilizing an enhanced exergy approach [83]. The dynamic simulation occurred on July 15, April 15, and October 15, from 8 a.m. to 5 p.m. during spring, summer, and autumn [84]. Advanced exergy assessment is utilized for the first time to provide direction and guidance for enhancing the practical implementation of an ejector based double-efficient CO₂ cooling and heating system. In cooling mode, 74.56 % of the total exergy destruction is endogenous, signifying limited interaction among the parts [85]. A proposed energy system integrates biomass, solar energy, and a two-stage ORC, with analyses conducted on conventional and advanced exergy, as well as exergoeconomic and exergoenvironmental factors [30]. Furthermore, both conventional and advanced exergy analyses are conducted on a double absorption heat transformer system that utilizes Substandard industrial waste heat for water treatment. Results from the advanced exergy assessment indicate that, for the suggested application, just 16 % of the total exergy degradation rate (25.48 kW) can be avoided [86]. A novel green hybrid multi-generation system utilizing CAES was introduced in the study. The ORC and RO desalination system designed to capture waste heat and deliver both potable water and domestic hot water (DHW). This system has the capacity to produce 37.48 MWh of electricity, generate 19.42 kg/s of drinkable water, and provide 2.67 kg/s of DHW [87]. The LNG maintained at -162 °C requires regasification. To effectively utilize the

significant amounts of thermal energy emitted during the regasification of LNG while minimizing investment, a multistage Rankine cycle system integrated with hydrate energy storage was developed. The Aspen HYSYS V11 software was utilized to design the LNG cold energy recovery process [88]. A study presented findings from a radial turbine-equipped RBC utilizing natural air (R-729), a safe fluid with no ozone depletion potential (ODP) and no emission capacity. The cycle coefficient of performance (COP) was evaluated at a design operating point, and an exergy assessment was conducted at 173K [38]. The thermodynamic study was effectively conducted for an ocean thermal engineering conversion system. The thermodynamic model established in this study, with $T_{ws} = 40.82\text{ }^{\circ}\text{C}$ and $T_{cs} = 23\text{ }^{\circ}\text{C}$, $\Delta T_{pinch} = 3$, calculates an efficiency of 2.05 % [89]. A water-to-water, segmental baffled shell-and-tube heat exchanger was selected for analysis and developed utilizing the Kern approach in the research, according to advanced exergy analysis 97.5 % of the overall exergy degradation is of the endogenous avoidable variety [90]. Numerous studies examine just one or two of the standard or advanced exergy, exergo economic, and exergo environmental [91,92]. Comprehensive exergy and exergo-economic evaluation of an integrated CO₂ capture and storage (CCS) and waste heat recovery system, measuring component exergy destruction and cost rates for optimization. An advanced exergy study demonstrates that 43.60 % of exergy degradation is preventable, requiring system modifications. External and endogenous exergy destruction optimizes distillation columns, evaporators, and turbines. The exergo-economic study also recommends reducing exergy destruction and capital expenditures, primarily for compressors and turbines with high investment costs. Advanced exergy and exergo-economic analysis can find energy system inefficiencies and improve them [93]. As individuals increasingly prioritize environmental issues, the impact on the environment has become as crucial concerning the design and optimization of energy systems as costs. Khoshgoftar Manesh et al. [94] conducted an evaluation of the combined biomass gasification cycle using both conventional and advanced 3E analyses, investigating the possibility for improvement in every component of system. A comprehensive thermodynamic and economic assessment of a 10 MW liquid carbon dioxide energy storage (LCES) system. They assessed the system performance using classic and advanced exergy/exergo-economic analytic methods. The approach introduced new notions like unavoidable/avoidable exergy destruction, endogenous/exogenous exergy degradation, and investment expenditure cost evaluations. Analyzing component interactions and detecting inefficiencies helps these advanced technologies find system improvements. Conventional investigation showed that the compressor greatly affected exergy destruction. However, sophisticated research has shown that the expander is the most critical component for enhancement. Sophisticated exergy approaches optimize energy systems and reduce costs, which can improve compressed gas energy storage systems [95]. Advanced and conventional exergy-economic assessments treat energy system inefficiencies and improvement potential differently. Conventional analysis identifies components with the most considerable exergy degradation but cannot identify preventable losses, limiting optimization. In contrast, advanced exergy analysis differentiates endogenous and exogenous losses for more precise improvement techniques. An enhanced examination of an AA-CAES system showed that 40 % exergy destruction may be avoided, providing significant optimization prospects. Advanced approaches better analyze exergy efficiency and cost reductions, exhibiting 77 % efficiency and 3.55 % cost savings compared to 54.51 % in conventional analysis. These methods identify critical components like tri-generation air compressors and examine environmental implications, promoting sustainability. Advanced exergy-economic analysis is essential for energy system evaluation and improvement as efficiency and sustainability requirements rise. Thermodynamic systems, such as Advanced Adiabatic Compressed Air Energy Storage (AA-CAES), must consider both thermodynamic and economic factors for effective design and implementation. Exergo-economic analysis integrates these elements, evaluating system

efficiency while accounting for costs related to maintenance, investment, and irreversibility. Conventional methods can identify inefficiencies and economic losses, while advanced techniques offer deeper insights into exergy destruction and potential cost savings. This study applies both traditional and advanced exergy/exergo-economic analyses, utilizing tools like parametric analysis and modified total unit product cost, to optimize AA-CAES systems. The findings highlight critical areas for reducing costs and improving exergy efficiency, underscoring the significance of advanced exergoeconomic analysis in enhancing the technical and economic performance of sustainable energy storage technologies [96]. Moreover, the review of investigated works is presented in [Appendix 1](#).

7. Conclusion

This review covers the detailed and complex methods of advanced exergy analysis, advanced exergoeconomic analysis, and advanced exergo-environmental analysis. These cutting-edge techniques expand the conventional framework for exergy analysis, providing deeper insights into the effectiveness of energy systems and their wider economic and environmental implications. Advanced exergy analysis distinguishes between avoidable and unavoidable exergy destruction, thereby enhancing our comprehension of energy systems. This differentiation enables the identification of particular areas that can be improved, focusing on operational or technological adjustments that have the potential to greatly increase system efficiency. This research lays the foundation for future technological advancements by concentrating on preventable exergy destruction, which directs efforts to minimize exergy losses and maximize system performance. The integration of economic aspects through advanced exergoeconomic analysis makes it feasible to evaluate the financial effects of inefficiencies in energy systems. This approach particularly aids in maximizing the economic performance of energy systems and ensures the allocation of funds towards the most significant enhancements by distinguishing between avoidable and inevitable expenses. Through the incorporation of environmental factors, exergo-environmental analysis creates This strategy links energy inefficiencies to environmental damage, promoting the creation of more sustainable energy systems and directing efforts to lessen environmental effects while preserving or enhancing energy efficiency. When used together, these advanced methodologies make up a complete framework for making decisions. Advanced exergy analysis finds thermodynamic inefficiencies, advanced exergoeconomic analysis links these inefficiencies to costs, and exergo-environmental analysis links them to environmental burdens. When used together, they give a complete picture of how well an energy system works, which helps with sustainable design, policy making, and investment planning. In summary, a major development in the field is the use of advanced exergy, advanced exergoeconomic, and advanced exergo-environmental evaluations in the assessment of energy systems. These approaches enable more informed decision-making and promote innovation in the design and optimization of energy systems by offering a thorough grasp of exergy inefficiencies, their financial implications, and their environmental effects. These cutting-edge analytical techniques will be vital in determining the future of energy systems and guaranteeing their sustainability and efficiency as the world's energy demand rises and environmental concerns grow.

CRedit authorship contribution statement

M. Fallah: Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Z. Moham-madi:** Writing – original draft, Supervision, Resources, Investigation. **S. Allahyari:** Writing – original draft, Investigation. **SH. Rahimi:** Writing – original draft, Investigation. **M. Fathi:** Writing – original draft, Investigation. **Z. Hoseini Tabar:** Writing – original draft, Investigation. **S.M.S. Mahmoudi:** Writing – review & editing, Supervision,

Methodology.

interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of competing interest

The authors declare that they have no known competing financial

Appendix 1. Review of performed studies

Authors	Year	Investigated system	Conventional analysis				Advanced analysis									Advanced exergy methods
			Energy	Exergy	Econ	Enviro	Exergy			Economical			Environmental			
							1st*	2nd*	Mex*	1st	2nd	Mex	1st	2nd	Mex	
Morosuk et al. [7]	2007	ARM	✓	✓			✓	✓								Thermodynamic cycle
Morosuk et al. [40]	2009	Refrigeration machines	✓	✓			✓	✓								Thermodynamic cycle
Morosuk et al. [15]	2009	The simple gas-turbine power system	✓	✓			✓	✓								Exergy balance method/Hybrid method
Petrakopoulou et al. [17]	2012	A combined cycle power plant	✓	✓			✓	✓								Thermodynamic cycle
Soltani et al. [42]	2013	An externally-fired combined-cycle power plant	✓	✓			✓	✓								Engineering method
Fallah et al. [25]	2016	A Kalina cycle	✓	✓			✓	✓								Thermodynamic cycle
Fallah et al. [43]	2016	Four gas turbine systems	✓	✓			✓	✓								The engineering method and the thermodynamic cycle method
Fallah et al. [44]	2017	Anode 284 recirculation SOFC system	✓	✓			✓	✓								engineering method
Mohammadi et al. [30]	2019	Recompression SCO ₂ cycle	✓	✓			✓	✓								Thermodynamic cycle
Ochoa et al. [46]	2020	A waste heat recovery system with a RORC	✓	✓		✓	✓						✓			
Mohammadi et al. [48]	2021	The S-CO ₂ recompression cycle	✓	✓	✓		✓	✓		✓	✓					thermodynamic cycle
Gullo et al. [49]	2021	Transcritical R744 booster supermarket refrigeration	✓	✓			✓	✓								
Zhao et al. [50]	2021	Joule-Brayton PTES	✓	✓			✓	✓								thermodynamic cycle
Song et al. [52]	2021	SOFC	✓	✓			✓	✓								engineering
Liu et al. [53]	2021	LCES	✓	✓			✓	✓								
Caglayan et al. [13]	2021	COGEN for ceramic industry	✓	✓			✓	✓								
Wu et al. [54]	2021	Coal-to-SNG	✓	✓			✓									
Muhammad et al. [55]	2021	CO ₂ pressurization	✓	✓			✓	✓								
Ucar et al. [56]	2021	A condensed combi boiler	✓	✓			✓	✓								engineering
Zhang et al. [57]	2021	LT-CCES	✓	✓			✓	✓								thermodynamic cycle
Wang et al. [58]	2021	DORC	✓	✓			✓	✓								thermodynamic cycle
Liu et al. [59]	2021	The EJE + TES	✓	✓			✓	✓								thermodynamic cycle
Wang et al. [16]	2021	The ORC and the ARC driven by waste heat with the CCS	✓	✓	✓		✓	✓		✓	✓					
Mohammadi et al. [23]	2022	The combined super critical carbon dioxide and organic Rankine cycle (SeCO ₂ /ORC)	✓	✓			✓	✓	✓							Thermodynamic cycle
Mohammadi et al. [37]	2022	An ammonia-water combined power and cooling cycle	✓	✓			✓	✓								thermodynamic cycle
Mohammadi et al. [26]	2022	The combined GT-MHR/ORC	✓	✓			✓	✓								thermodynamic cycle
Ustaoglu et al. [60]	2022	ORC-CHP	✓	✓			✓									
Ayub et al. [61]	2022	SOFC based tri-generation process	✓	✓	✓		✓	✓		✓	✓					
Zheng et al. [62]	2022	CO ₂ two-phase ejector	✓	✓			✓	✓								
Montazerinejad et al. [63]	2022	PEMFC and ORC	✓	✓			✓	✓								

(continued on next page)

(continued)

Authors	Year	Investigated system	Conventional analysis				Advanced analysis									Advanced exergy methods
			Energy	Exergy	Econ	Enviro	Exergy			Economical			Environmental			
							1st*	2nd*	Mex*	1st	2nd	Mex	1st	2nd	Mex	
Ozen et al. [64]	2022	A novel combined power system using the cold energy of LNG	✓	✓	✓		✓	✓		✓	✓					Modified Productive Structure Analysis
Hamed et al. [31]	2022	An existing industrial olefin plant	✓	✓	✓	✓	✓	✓		✓				✓		Engineering method
Güleri yüz et al. [65]	2022	AA-CAES	✓	✓	✓		✓	✓		✓	✓					
Gürbüz et al. [66]	2022	GPP	✓	✓		✓	✓	✓					✓	✓		Engineering method
Nourpour et al. [67]	2022	An integrated combined cycle including GTs, SOFC, SORCs	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓		Engineering method
Mohammadi et al. [68]	2023	Six layouts of S- CO ₂ cycles	✓	✓				✓	✓							thermodynamic cycle
Mohammadi et al. [69]	2023	Double-flash cycle	✓	✓												thermodynamic cycle
Sheng et al. [70]	2023	The oil and gas processing plant	✓	✓				✓	✓							thermodynamic cycle
Tian et al. [71]	2023	The cryogenic recuperative ORC	✓	✓				✓	✓							thermodynamic cycle
Li et al. [72]	2023	An ORFC	✓	✓	✓			✓	✓		✓	✓				
Caliskan et al. [73]	2023	An ejector-based CO ₂ transcritical CCHP	✓	✓				✓	✓	✓	✓	✓	✓			thermodynamic cycle
Zheng et al. [74]	2023	SCESHs	✓	✓				✓	✓							thermodynamic cycle
Esmaeilion et al. [75]	2023	MED	✓	✓				✓	✓		✓	✓				thermodynamic cycle
Güleri yüz et al. [97]	2023	AA-CAES	✓	✓	✓			✓	✓		✓	✓				
Liu et al. [76]	2023	WSH-MES	✓	✓				✓	✓							
Liu et al. [77]	2023	CtEG process	✓	✓				✓	✓							
Liu et al. [78]	2023	LT-PEMFC and HE-ABS	✓	✓				✓	✓							
Esmaeilion et al. [75]	2023	MED	✓	✓	✓			✓	✓		✓	✓				thermodynamic cycle
Khan et al. [79]	2023	Single slop solar still	✓	✓				✓			✓			✓		
Chi et al. [80]	2023	NH ₃ /CO ₂ CRSE	✓	✓				✓	✓							
Xiao et al. [81]	2024	Tri-generation systems driven by SOFC and GT	✓	✓	✓	✓			✓			✓			✓	
Liu et al. [82]	2024	A <i>trans</i> -critical CO ₂ energy storage system	✓	✓				✓	✓							thermodynamic cycle
Ma et al. [83]	2024	An air source AHTHP	✓	✓				✓	✓							
Hajizadeh et al. [84]	2024	An indirect solar dryer	✓	✓				✓	✓							
Wang et al. [85]	2024	DECHS	✓	✓				✓	✓	✓						
Qi et al. [27]	2024	An ORC integrated with solar and biomass	✓	✓	✓	✓		✓	✓		✓	✓		✓	✓	
Mehrjouie et al. [86]	2024	DAHT	✓	✓				✓	✓							thermodynamic cycle
Bushehri et al. [87]	2024	CAES-ORC-RO	✓	✓	✓			✓	✓		✓	✓				
Liu et al. [88]	2024	A multi-stage Rankine cycle system	✓	✓	✓			✓	✓		✓	✓				
Serrano et al. [38]	2024	RBC	✓	✓				✓	✓							Thermodynamic cycle method
Kim et al. [89]	2024	The OTEC	✓	✓				✓	✓							
Prajapati et al. [90]	2024	The shell and tube heat exchanger						✓	✓							
Ozen et al. [93]	2024	AA-CAES system with a power output	✓	✓	✓			✓	✓		✓	✓		✓	✓	
Khandelwal et al. [95]	2024	ISCC	✓	✓	✓	✓										
Yinlong et al. [96]	2024	EARC	✓	✓				✓			✓			✓		

Data availability

No data was used for the research described in the article.

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